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Complete List of Authors:	Sheng, Yifei; North China University of Science and Technology, School of Clinical Medicine Liu, Chengjiang; Anhui Medical University, Department of General Medicine Hao, Siyu; North China University of Science and Technology, School of pharmacy Jia, Xiaodong; North China University of Science and Technology, School of pharmacy Feng, Xinyi; North China University of Science and Technology, School of pharmacy Yang, Qian; North China University of Science and Technology, School of Clinical Medicine
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Gut Microbial Diversity, its role in Type 2 diabetes and Computational Based Metagenomic Methods for Gut Microbial Characterization

Yifei Sheng ^{1,*}, Chengjiang Liu ², Siyu Hao ³, Xiaodong Jia ³, Xinyi Feng ³, Zhijia Fu ¹

¹ School of Clinical Medicine, North China University of Science and Technology, Hebei province, 063210, China

² Department of General Medicine, Affiliated Anqing First People's Hospital of Anhui Medical University, 246000, China

³ School of Pharmacy, North China University of Science and Technology, Hebei province, 063210, China

***Corresponding author:** Yifei Sheng, School of Clinical Medicine, North China University of Science and Technology, Hebei province, 063210, China, Email: shengyifei@stu.ncst.edu.cn

Abstract: The intestinal microbiome plays a pivotal role in human health and disease, with emerging data underscoring its contribution to the pathogenesis of type 2 diabetes (T2D). The purpose of this literature review is to offer a comprehensive examination of intestinal microbial diversity, its association with T2D, and the computational metagenomic methodologies employed for profiling intestinal microorganisms. Intestinal microbial diversity refers to the array of microbes residing in the gastrointestinal tract, which is influenced by various factors such as diet, lifestyle, and host genetics. Changes in the composition and functionality of intestinal microorganisms have been linked to the onset of T2D, indicating a potential involvement of the intestinal microbiome in the development of the disease. Recent progress in computational metagenomic techniques has transformed the exploration of the intestinal microbiome, facilitating the high-throughput analysis of microbial communities. Approaches such as sequencing of the 16S rRNA gene and shotgun metagenomics empower scientists to delineate the composition and functionality of the intestinal microbiome in states of health and disease. Understanding the importance of intestinal microbial diversity in T2D and utilizing computational metagenomic methodologies are crucial advancements in the pursuit of personalized strategies for preventing and managing the disease. This review underscores the significance of intestinal microbiome research in enhancing our comprehension of T2D and its treatment.

Keywords: Gut microbiota; Microbial diversity; Type 2 diabetes; Metagenomic methods; Computational analysis

1.0 Introduction

The gut microbiota consists of intricate ecosystems including billions of bacteria, encompassing a minimum of 1000 distinct species [1]. The majority of gastrointestinal microbiota is comprised of bacteria, although archaea, viruses, fungi, and protists are also present [2]. The connection between the host and the intestinal microbiota is crucial and plays a significant influence on the host's physiological system. An increase in the diversity and quantity of bacteria and fungi in the gut microbiota serves as a key indication of dysbiosis, which can result from changes in the population of bacteria and fungi, particularly those linked to dysfunction or pathogenesis [3]. There is a correlation between gut microbiota dysbiosis and illnesses affecting the cardiovascular, neurological, immunological, and metabolic systems [4]. Various studies have shown that unregulated insulin sensitivity and reduced glucose tolerance lead to a rise in body weight [5], metabolic syndrome, and the development of type 2 diabetes (T2D) through heightened glucose intolerance [6,7]. Several researches have shown that gut microorganisms can impact chronic systemic inflammation, mostly through endotoxemia, which is the release of endotoxins after bacterial death [8]. There is a limited amount of research that has investigated the role of the gut microbiota in the development and advancement of diabetes-related complications. While the connection between intestinal microbiota and diabetes is still under investigation, some research has explored the pathophysiology of diabetes [9].

In research investigating the correlation between the gut microbiota and T2D, it is crucial to include the impact of drugs [10-16]. T2D is a complex metabolic disorder that includes many aspects and impacts a substantial global population. The relationship between gut microbiota and the development and progression of T2D has received significant interest due to its association with food and digestion. The discovery of various profiles of multiple bacteria associated with the disease in individuals of different ages and ethnicities further complicates the identification of the specific microbes responsible for the sickness [17-19]. The gut microbiota has the capacity to impact cardiovascular health through various pathways. One such pathway involves the production of metabolites such as trimethylamine N-oxide (TMAO) by specific gut bacteria, a factor known to be correlated with an elevated susceptibility to cardiovascular diseases (CVD). Conversely, certain gut bacteria have the ability to generate short-chain fatty acids (SCFAs) that possess anti-inflammatory properties and could potentially offer protection against CVD [20]. The role of the gut microbiota extends to the modulation of the neurological system via the gut-brain axis. Through this axis, the gut microbiota can synthesize neurotransmitters like serotonin, dopamine, and gamma-aminobutyric acid (GABA) that have implications for mood, behavior, and cognitive functions. Furthermore, an association has been established between gut dysbiosis (an

imbalance in gut microbiota) and neurological conditions such as depression, anxiety, autism spectrum disorders, as well as neurodegenerative ailments like Alzheimer's and Parkinson's [21]. Alterations in the intestinal microbiota configuration, specifically a reduction in microbial variety, have been detected in persons with colorectal cancer (CRC). Particular bacteria, including *Fusobacterium nucleatum* and *Bacteroides fragilis*, have been correlated with the development of CRC. On the other hand, certain beneficial bacteria, such as *Faecalibacterium* and *Roseburia*, may provide protection against CRC by producing anti-inflammatory compounds [22].

The crucial function of gut bacteria in the development and progression of T2D including other disease is currently being investigated and requires further study in this field. It is also needful to investigate microbial diversity in the T2D patients to understand the role of microbes in disease development. Metagenomics is a culture independent method of identification of gut microbiota in the environmental or patient samples [23]. Metagenomics is use to study gut microbial community of diabetes individuals. Microbes may be studied at the genus and species level using both culture-independent and culture-dependent methods of microbial diversity study. Furthermore, metatranscriptomics, metabolomics, and metaproteomics offer insight into the functional significance of genes and enzymes within the microbial population of both healthy and diabetic patients [24-26].

Research on the gut microbiota generates huge amounts of data; yet, the methods used to analyse this data are always evolving and enhancing [27]. The classification, data processing, and modelling of the gut microbiota are crucial steps in the discovery of microbial species and new genes. However, it is imperative to include them in the conventional in vitro and in vivo modelling methods as well. Metagenomics is a methodology that involves analyzing the makeup of microbiome groups to identify species and reduce reliance on culture-based methods [28]. Advancements in next-generation sequencing (NGS) and computational methods enable the exploration of the structural composition and functional abilities of the microbiome [29].

This review is focused on the gut microbial analysis using metagenomic based tools. This review is also focused on the role of gut microbiota and microbial dysbiosis in the T2D patients. Authors have also described several methods of target sequencing and whole-genome metagenomic sequencing.

2.0 Gut microbiota and their functions

The gut microbiota is mostly composed of anaerobic bacteria, rather than facultative anaerobic bacteria, which can flourish in both aerobic and anaerobic conditions [30]. The dominant microbiota is characterized by a restricted number of taxa, specifically the Bacteroidetes and the

Firmicutes. Several studies have been reported that human gut contains around one thousand types of bacterial phylotypes [31, 32].

The human gastrointestinal tract (GIT) contains different type of bacteria depending on their abundance. Moreover, the stomach and duodenum contain bacteria at a density ranging from 10 to 10^3 per gram. In the small intestine, these germs are found at a density between 10^4 and 10^7 per gram [33]. Conversely, the large intestine contains microorganisms at a density ranging from 10^{11} to 10^{12} per gram. Significantly, several bacterial phyla are more abundant in the small intestine and colon, resulting in diversity among the microbial population in these areas. Frank et al. [34] discovered that healthy individuals have a higher prevalence of Firmicutes bacteria in the small intestine than in the colon. This finding suggests that the bacterial population in the human gastrointestinal tract plays a significant role in human metabolism and the development of diseases [34].

Gut bacteria can produce a wide range of vitamins as well as both necessary and nonessential amino acids [35]. In addition, the microbiome has a crucial function in aiding the breakdown of carbohydrates that cannot be digested, such as resistant starches, cellulose, hemicellulose, pectins, gums, large polysaccharides, oligosaccharides, sugars, alcohols, and mucins produced by the hosts [36, 37]. As a result, the host may harness energy from the bacteria and assimilate substrates. Conversely, the bacteria can flourish and proliferate as a result of the energy and nutrients supplied to them. The colon produces many energy sources through the process of carbohydrate metabolism [38].

2.1 Host immune-system development and protection

The immune system and the environment are directly linked in the intestinal epithelium, which serves as the main contact between them. The host immune system and the intestinal microbiota and its metabolites engage in continuous and dynamic interaction, which influences the development and functioning of the host immune system. Bacteria are widely recognized for their crucial contribution to the development of the gut-mucosal immune system throughout the early stages, both in terms of physical and biological aspects. Furthermore, they maintain a vital function in their activity into the later stages of life [38]. The intestinal epithelial cells communicate with the innate immune system to prevent the invasion of pathogens by using receptors that can identify and attach to particular chemicals associated with germs. By eliciting this response, the host is prompted to activate its defensive mechanisms, resulting in the release of protective peptides, cytokines, and white blood cells that will safeguard it against invading pathogens. An immune reaction to a harmful microorganism can lead to defence against harmless bacteria, while a defensive response to harmless bacteria can cause the host's cells to die. Furthermore, research has

demonstrated that being exposed to gut bacteria can effectively mitigate allergies, which refer to an exaggerated immune reaction to an innocuous antigen [39]. The hypotheses propose that the bacteria residing in the gastrointestinal tract activate the immune system and instruct it to react appropriately to various antigens by shaping the intestinal microbiota for this purpose. During the initial stages of development, a modified arrangement of the gut microbiota might lead to an immature immune system that is susceptible to exaggerated responses towards antigens. This tendency to overreact to antigens is a frequently observed phenomenon [40].

2.2 Role of gut microbiota in T2D

The previous study indicated a possible role of gut bacteria in the development of metabolic diseases such as T2D [41]. Subsequent research has observed that mice with obesity display an increased variety of Firmicutes in their gut microbiota. Conversely, mice who are slim exhibit a higher level of variety in their Bac-teroidetes population [42]. Recent studies have shown that an individual's body weight has an impact on the composition of their gut flora. Research examining obese patients found that their gut microbiota exhibited variations compared to persons with a normal weight, and these differences were influenced by changes in body weight [43]. However, subsequent investigations have been unable to show a reliable connection between obesity and the ratio of Firmicutes to Bacteroidetes, both at the phylum and species levels [44].

Due to its genetic and environmental factors, this issue has become a major concern in public health worldwide [45]. During the investigation, primarily utilize genome-wide association studies to determine genetic elements within the genome and identify the underlying genetic factors that contribute to the recent inquiry. It is discovered that the composition of the intestinal microbiota may also have a significant impact on the development of T2D. The gut microbiota of individuals with T2D showed a decrease in the abundance of bacteria that generate butyrate, a compound that may contribute to metabolic well-being, compared to individuals without the illness [46]. Hence, it may be hypothesized that functional dysbiosis may have a stronger association with the occurrence of T2D than the particular microorganisms implicated. A wide range of opportunistic infections experienced a significant rise in their prevalence. Nevertheless, the prevalence of these diseases exhibited significant fluctuations [47].

The gut microbiome pay important role in the nutrients obtaining and storing energy. In addition, they produce exometabolites that might potentially affect the condition of the host's metabolic processes. Short-chain fatty acids (SCFAs) are external metabolites that directly impact metabolic processes [48, 49]. Several studies have shown that SCFAs might cause metabolic disorders such as obesity and T2D [48, 50]. It has been obserbed that individuals, including adults and children, have higher levels of SCFAs in their faeces, specifically butyrate and propionate, in comparison

to lean individuals [51]. The prevalence and dispersion of particular bacterial phyla might be associated with fluctuations in the concentrations and proportions of distinct SCFAs [52].

3.0 Gut microbiota dysbiosis in T2D patients

There is a direct connection between the gut microbiota and obesity, non-alcoholic fatty liver disease (NAFLD), insulin resistance, and chronic inflammation [53,54], all leading to the development of T2D. A form of gut bacteria known as *Oscillospira* was the subject of a study that was carried out on Italian patients suffering from NAFLD. The researchers found that these individuals had lower amounts of the bacteria than the healthy control group. In addition, the individuals with NAFL showed more significant amounts of both 2-butanone and 1-pentanol in their systems. The findings indicate that there may be a link between the number of intestinal bacteria and liver steatosis, underlining the possible function that gut microbiota plays in this illness. In addition, the results imply that there may be a correlation between the abundance of intestinal bacteria and obesity. Del Chierico et al. [55] describe how the investigation was carried out using targeted metagenomics and metabolomics approaches. The researchers found that feeding Syrian hamsters a diet that was rich in both fat and sugar might result in a change in the microbiota makeup of the gastrointestinal tract. This alteration may result in the development of dyslipidemia as well as hepatomegaly [56]. There is a significant association between the levels of fasting cholesterol and the presence of *Ruminiclostridium* and *Tyzzereella* in the gut flora. There is also a positive correlation between fasting triglyceride levels and the presence of *Tyzzereella* and *Ruminococceace* NK4A214 group [57]. Based on these findings, there appears to be a significant connection between the microbiota in the gut and a group of metabolic illnesses. It is important to note that the gut microbiota composition has been demonstrated to be diabetic in humans and animal models, suggesting that it may play a significant role in the aetiology of T2D. The diabetic-induced complications and bacteria dysbiosis in diabetic patients are shown in Figure 1.

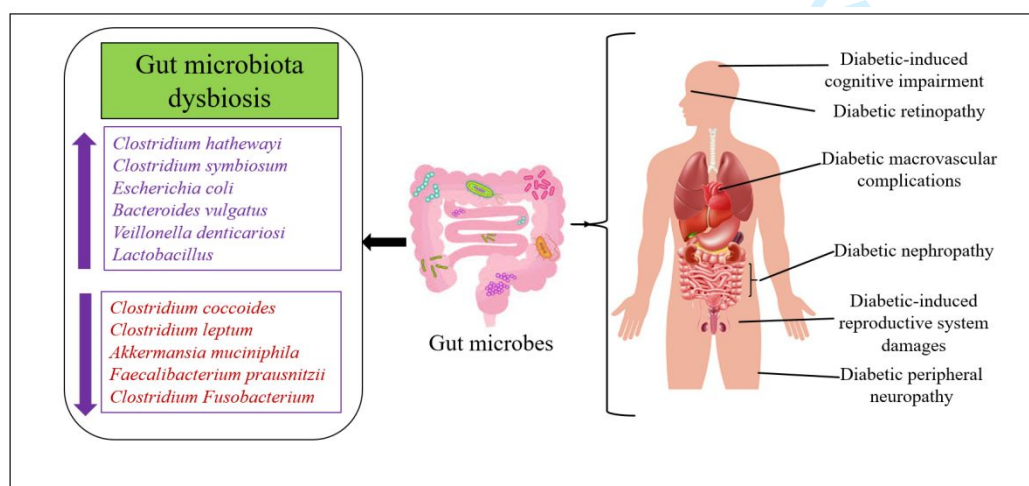


Figure 1. This diagram depicts the gut microbiota dysbiosis and diabetic-related complications

An imbalance in the gut microbiota can arise not only in individuals who have T2D but also in animal models such as mice, cats, and zebrafish, according to research conducted [58, 59]. In addition, the gut microbiota has been shown to have a close relation to several diabetes-related problems, including diabetic nephropathy, diabetes-related cognitive impairment, diabetic retinopathy, and diabetic peripheral neuropathy [60].

There have been a significant number of reports of people suffering from dysbiotic gut microbiota who were diagnosed with T2D. Qin et al. [61] investigated that patients with T2D had a rise in several harmful bacteria, such as *Escherichia coli*, *Clostridium Hathaway*, and *Clostridium symbiosum*. In contrast, healthy controls exhibited a significant concentration of butyrate-producing bacteria. In addition, a second investigation indicated that in European women with T2D, four *Lactobacillus* species were found to have a higher abundance. Still, five *Clostridium* species were found to have a lower abundance [62]. This was compared to healthy patients who had standard glucose tolerance.

Consequently, species of *Lactobacillus* had a positive correlation with fasting glucose levels and glycosylated haemoglobin levels (HbA1c). In contrast, species of *Clostridium* had a negative correlation with fasting glucose levels, HbA1c levels, and plasma triglyceride levels. According to these findings, certain bacterial taxonomic groups may play a part in the progression of T2D. According to Chen et al. [63], new T2D patients exhibited considerably greater amounts of *Lactobacillus* and lower levels of *Clostridium coccoides* and *Clostridium leptum*, which indicated that the number of new T2D patients had higher levels of *Lactobacillus* in their bodies. In addition, Shih et al. [64] investigated the features of the microbiota in the intestinal tracts of persons with refractory diabetes mellitus (RT2DM) who were already being treated with metformin but whose HbA1c levels continued to rise by at least 8% [64]. *Bacteroides vulgatus* and *Veillonella denticariosi* exhibited a greater abundance in the RT2D patients than in the T2D controls. In contrast, *Akkermansia muciniphila* and *Fusobacterium* showed a lower abundance in the RT2D patients than the T2D controls. *Bacteroides vulgatus* and *Veillonella denticariosi* showed a higher abundance in the RT2D patients than in the T2D controls. The bacterium known as *A. muciniphila* was the only one that had an inverse association with the amount of haemoglobin HbA1c [65]. The bacterial diversity in the gut of diabetic patients and major metagenomic techniques for bacteria identification are shown in Table 1.

Table 1. Microbial diversity in T2D patients and techniques for microbial identification

Patient type	Metagenomic techniques	Observation	References
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		Several biomarkers have been identified for T2D, including <i>Acidaminococcales</i> , <i>Bacteroides</i> , <i>Plebeius</i> , and <i>Phascolarctobacterium</i> sp. These biomarkers have the potential to significantly improve T2D treatment. Moreover, correctly understanding the metabolic and environmental factors that influence the presence of these biomarkers could ultimately lead to better preventive measures and management strategies for T2D. However, it is important to note that further scientific research is needed to understand the significance of these biomarkers in T2D fully. There is a higher frequency of certain bacterial families and orders, including	
T2D	Metagenomic shotgun sequencing	<i>Bifidobacteriaceae</i> , <i>Coriobacteriaceae</i> , <i>Actinobacteria</i> , <i>Caulobacteraceae</i> , <i>Phyllobacteriaceae</i> , <i>Rhizobiales</i> , and <i>Burkholderiaceae</i> .	[66, 67]
T2D	16S rDNA gene sequencing by Illumina MiSeq platform	<i>Lactobacillus</i> levels increased while <i>Clostridium coccoides</i> and <i>Clostridium leptum</i> levels decreased in newly diagnosed T2D patients.	[68]
T2D	16S rRNA sequencing	Patients with T2D had a higher abundance of <i>Bacteroides vulgatus</i> and <i>Veillonella denticariosi</i> , and a lower abundance of <i>Akkermansia muciniphila</i> and <i>Fusobacterium</i> .	[69]
Refractory T2D	16S rRNA sequencing		[70]

Prediabetes refers to a situation where people have elevated blood glucose levels above normal levels. Still, they do not exceed the thresholds indicating diabetes [71]. It has been established that individuals with prediabetes are more likely to acquire T2D than people without prediabetes. Prediabetes is a condition in which blood glucose levels are higher than normal but are still below the threshold levels that indicate T2D. Some intriguing discoveries about prediabetes patients having abnormal gut microbiotas [72, 73] found a substantial difference in the gut microbiota of patients with prediabetes compared with those with standard glucose control. Specifically, there was a drop in the abundance of the *Clostridium* genus and the *Amoeba muciniphila*. According to the findings of a study that was carried out by Zhong et al. [74], there was not a discernible difference in the levels of microbial gene-based richness that existed between treatment-naïve Chinese patients who were diagnosed with T2D and prediabetes and Chinese individuals who had standard glucose tolerance. Despite this, individuals with prediabetes had significantly lower

abundances of metagenomic linkage groups (MLGs) compared to individuals with standard glucose tolerance. However, higher abundances of MLGs from *Lactobioccus*, *Streptococcus salivarius*, and *Eggerthella* sp showed a correlation with the abundance of MLGs in the *Clostridia* class [74]. It was also shown that persons with prediabetes had a higher number of MLGs attributed to *E coli* than those with T2D who had not yet started therapy. The findings point to the possibility of complex interrelationships between the host and the gut microbiota and indicate that the gut microbiota may display varied characteristics at different phases of T2D [74].

4.0 Metagenomic-based characterization of gut microbiota

Pyrosequencing of the V4V5 region of 16S rRNA genes provided evidence that the diversity of the gut microbiota of diabetic patients was much lower than that of healthy persons [75]. Patients with T2D demonstrated a significant decrease in the number of bacteria that produce butyrate, such as *Bifidobacterium* and *Akkermansia*. In addition, a much larger number of *Dorea* were found in individuals with T2D, and these bacteria were found to be inversely associated with the amount of butyrate-producing bacteria. Researchers discovered that the genera *Ruminococcus*, *Fusobacterium*, and *Blautia* had a positive link with T2D. Still, the genera *Bifidobacterium*, *Bacteroides*, *Faecalibacterium*, *Akkermansia*, and *Roseburia* had a negative correlation with T2D [76].

4.1 Sample collection and DNA extraction

The first stage of the microbial identification of T2D patients is necessary to collect samples from T2D suffered patients. It is necessary that none of the participants received antibiotic therapy in the period immediately preceding the collection of the pieces (two months, to be exact). The specimens cryopreserve as soon as possible, and then DNA isolates from them using standard protocols [77, 78].

4.2 Metagenomic analysis of gut microbiota

Pyrosequencing of the V4V5 region of 16S rRNA genes revealed that the gut microbiota exhibited significantly reduced diversity compared to that of healthy individuals. Recent study focused on the identification of uncultured bacteria by NGS technology. Before investigating the functional involvement of microbial communities in the pathophysiology of disorders, it is imperative to accurately identify these communities with a high level of sensitivity and specificity. We can only assess the functional contribution of microbial communities to the development of illnesses by examining them [79, 80]. Ghaisas et al. [81] elucidate that there are two culture-free techniques available for studying the gut microbiota [81]. These investigations employ targeted and shotgun metagenomic sequencing techniques [82, 83]. Meta-genomics is a complex approach used to study microbial communities. It involves many steps, which are illustrated in Figure 2.

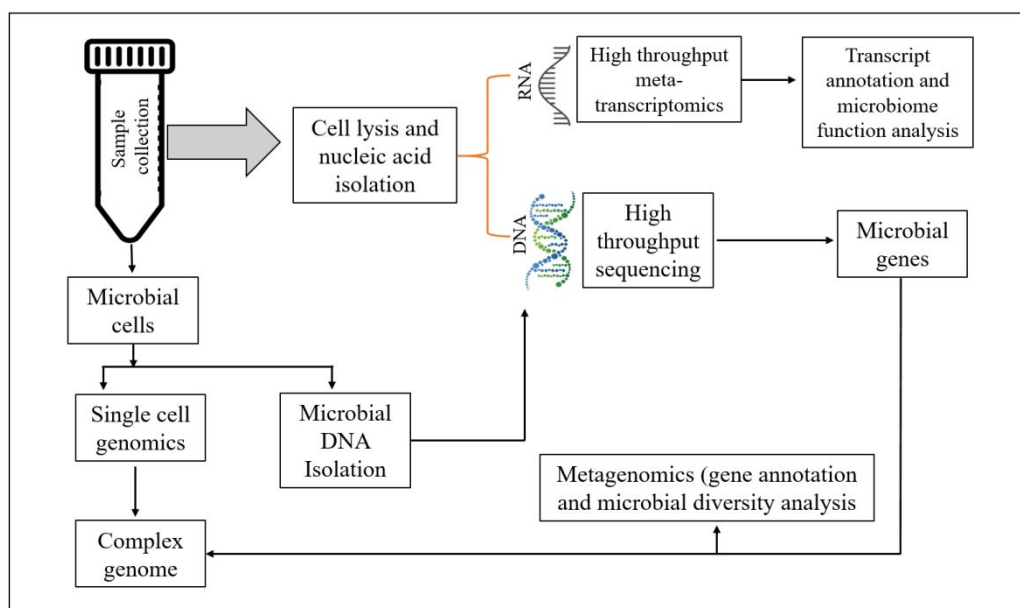


Figure 2. An overview of a metagenomic method for analyzing the gut microbiome

The stool sample analysis process involves extracting RNA from the sample and creating a cDNA library, digitized and accessible to researchers through platforms such as Illumina and 454 pyro-sequencing. The species present in the sample can be identified by clustering the sequences and comparing them with a reference database [84]. Although popular databases like RDP, SILVA, and GreenGenes can be used for this purpose, they may be outdated. To compare differences between samples or groups of samples, researchers can analyze the abundance of identified taxa using metrics like alpha diversity and beta diversity, principal coordinate analysis, and differential abundance on platforms like QIIME2. The same table can be used to infer metagenomic data, which may help predict the functional implications of differences in the microbiome composition [85].

4.2.1 Targeted sequencing

The 16S rRNA gene, a ubiquitous genetic component among bacterial and archaeal communities, has been universally recognized as the most important housekeeping gene [86]. Cryan et al. [87] demonstrated that sequencing 16S rRNA, a well-established, reliable, and relatively inexpensive method of determining the relative abundance of bacteria within gut microbiome that were sequenced using next-generation sequencing technologies. This method use for determining the relative abundance of bacteria in humans, mice, and insects. The relative abundance of microorganisms in the samples was determined by using this method. Targeted sequencing is one of the most common terms to describe this type of sequencing. Metagenomics sequencing using 16S rRNA would be more practical and more accessible in a laboratory setting when the inquiry focuses on the bacterial and archaeal communities in many control and diabetes patient samples [88].

The PCR is used to amplify the genetic sequence that is highly conserved. Based on their genetic linkages, the amplified sequences, referred to as amplicons, have been grouped into operational taxonomic units (OTU). This procedure was carried out on the amplified sequences [86]. The relative abundance that may be found in samples has been possible to be calculated because of this development. It is important to remember that other approaches have been utilized to define OTU, such as the de novo and closed-reference methods [89]. It is essential to highlight that the amplicons were recently inferred to be amplicon sequence variations (ASVs). To accomplish this, a variety of de-noising algorithms, such as Deblur and DADA2, as well as others, were applied [90].

4.2.2 Shotgun metagenomic sequencing

Shotgun metagenomic sequencing is more expensive than targeted sequencing. The results of Shotgun metagenomic sequencing are more accurate and informative [91]. It includes all of the microbial analysis present in a sample, which causes it to be more complicated. On the other hand, whole-genome sequencing can show taxa at the species level with a higher degree of accuracy and offers several benefits over 16S rRNA sequencing [92]. In particular, whole genome sequencing (WGS) broadens the detection range of bacterial diversity, improves the accuracy of species identification, assesses the functional potential of the microbiome, and discovers additional strains or mutations already present in samples [92].

The NGS library construction for RNA or DNA uses a procedural methodology that results in variations between studies. Regardless of whether a complete sample is being sequenced, this methodology entails the generation of infinitesimal reads ranging from 25 to 500 base pairs. This allows the identification of microorganisms that are either unknown or exist in minute quantities. Chiu et al. [93] reported that extensive bioinformatics preprocessing tools are required, including pruning, merging, assembly, scaffolding, and mapping tools [93]. Following the sequencing procedure, distinct sequences of the microbial components of the samples will be produced in fasta or fastq files, along with a mapping file that contains all the necessary metadata associated with the sample. It is said that these files serve as inputs to the subsequent identification of the species to which the sequences belong and the assignment of taxonomy to the sequences [94]. Using the term OTU as shown in Figure 3, it is possible to identify groups of similar sequences that have the potential to represent a distinct taxonomic classification based on these similarities.

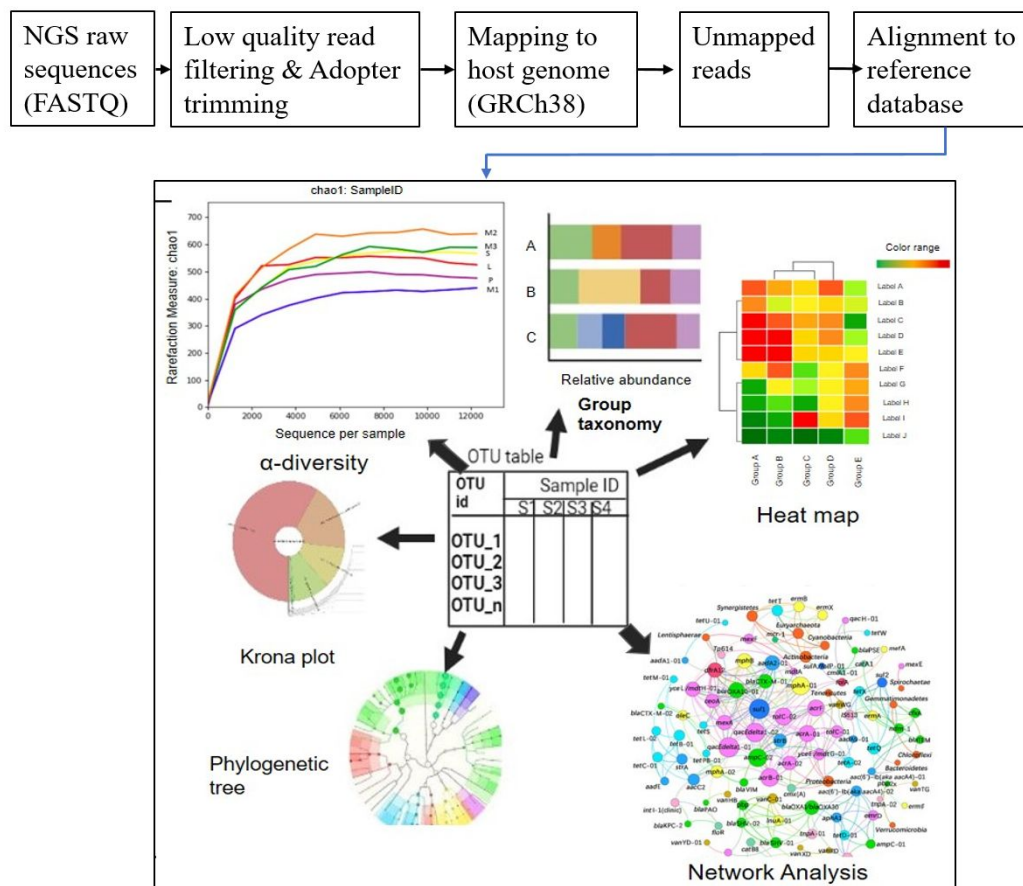


Figure 3. An overview of whole metagenome analysis of gut microbiota analysis

Although this method is not without faults, it is remarkably effective for clustering sequences with 97% similarity. Using phylogenetic alignment, one sequence is selected per OTU to represent its corresponding taxa. Numerous bioinformatics techniques and algorithms have been devised in the field of shotgun and 16S rRNA metagenomics, either as independent homology- and prediction-based methods or as components of more comprehensive workflows [95, 96].

Multiple investigations, utilising 16S rRNA analysis, have demonstrated a connection between the gut microbiota and general well-being [97-99]. An extensive examination of the gut metagenome, including whole genome sequencing (WGS), can enhance our understanding of the development of illnesses and allow for the discovery of new therapeutic targets. This occurs because there is a chance of discovering small genetic differences among species that cause changes in physical traits, ultimately resulting in the development of diseases. For instance, Whole Genome Sequencing (WGS) investigations carried out with *Citrobacter* spp. have shown that genetic differences within the species lead to changes in their observable characteristics and ability to adapt to different environments [100].

At present, the use of Illumina shotgun sequencing of stool samples is widely prevalent in the field of whole genome sequencing (WGS) studies of the gut microbiome. This is primarily because the gut harbours a multitude of diverse microbial species, thereby necessitating a thorough sequencing

process with a coverage of at least 20 times. The purpose of such extensive sequencing is to investigate and analyze individual communities within the gut microbiome that possess low abundance [100]. However, it is important to analyse of the substantial amount of WGS data, particularly in the form of short reads, poses significant challenges. This is needful because the gut microbiome is home to a wide range of bacterial species, ranging from hundreds to thousands, all of which exhibit varying levels of abundance. Complicating matters further, there is a lack of taxonomic identification available for the majority of these species, further exacerbating the analytical complexities [100, 101].

4.3 Metagenomic tools for gut microbiota analysis

The metagenomic data analysis can be performed by numerous software, web servers, and R packages which are currently accessible in the public domain. Some important metagenomic tools are given in Table 2.

Table 2. Metagenomic tools for gut microbiota analysis

Tools	Main function	Advantage	Platform	References
mothur	Microbial ecology analysis of Gut microbiota	This tool has the ability of fast process of large sequence data set	Linux, macOS, Windows	[102]
MEGAN	Taxonomic analysis and visualization.	This tool is applicable for the analysis of large metagenomic shotgun sequencing data	Windows, macOS	[103]
MetaPhinder	Identification of phages in metagenomic data	Phage identification in metagenomic datasets	Linux	[104]

		This tool has		
		important		
		application in		
		the generation		
MOCAT	Metagenomic	and assembly	Linux	[105]
	data analysis	of taxonomic		
		profiles and		
		assemble		
		metagenomes		
		QIIME is use		
		for Net-work		
		analysis of		
		meta-genomic		
		data and		
QIIME	Microbiome	histo-grams	Linux,	
	analysis	between	macOS,	[101, 106]
	pipeline	microbial	Windows	
		sample		
		diversity or		
		histograms		
		within the		
		sample.		
		This		
		computational		
		tool is		
		applicable for		
MetaPhlAn	Microbial	the faster	Linux,	
	composition	profiling of	macOS	[107]
	profiling	the microbial		
		composition		
		of microbial		
		communities		
		by clade-		

		specific		
		marker genes		
		High-		
	Strain-level	resolution		
	analysis of	strain		
ConStrains	metagenomic	identification	Linux	[108]
	data	from		
		metagenomes		
	Metagenomic	Efficient		
	data	analysis of		
MEBS	processing and	metagenomic	Linux	[109]
	analysis	sequencing		
		data		

We primarily concentrate on metagenomic software designed for the study of gut microbiota metagenomic data. There are methods available for analysing 16S rRNA marker gene sequencing data and whole-genome sequencing (WGS) data. Quantitative Insights into Microbial Ecology (QIIME) is a programme that focuses on analysing microbial diversity using 16S rRNA data. This tool enables users to allocate taxonomy for phylogenetic research, while also providing demultiplexing and quality screening for raw data obtained from Illumina or similar platforms. It is important to acknowledge that installing QIIME needs a specific level of proficiency in Linux and Windows systems. Additionally, it cannot do parallel processing during the OTU selecting step [101, 108]. Another significant software package is mothur, which is a versatile toolkit with a wide range of functions. These functions include the identification of operational taxonomic units (OTUs) and the comprehensive representation of alpha diversity within a specific sample, as well as beta diversity between different samples [102, 109].

RAMMCAP is a GUI-based program specifically developed for metagenomic sequence grouping and analysis. This tool possesses the capacity to effectively handle a substantial quantity of sequences within a significantly restricted timeframe, hence distinguishing it from other accessible tools and software alternatives. In addition, RAMMCAP incorporates a protein family annotation tool, enhancing its capability and use. In addition, RAMMCAP provides a distinctive metagenome comparison technique that relies on statistical analysis. This technique is likewise executed using a graphical user interface (GUI) [106]. Various methods are available for taxonomic binning when analyzing sequencing data generated from whole-genome sequencing (WGS) methodologies.

These methods commonly use the BLAST algorithm for species identification. An example of such a tool is the MEtaGenome ANalyzer (ME-GAN), which employs a BLAST search against a reference sequence database, such as the non-redundant sequence database from the NCBI NR database. The analysis yields results that are shown via a Graphical User Interface (GUI), therefore improving the user's overall experience. In addition, ME-GAN has the benefit of being capable of analyzing extensive datasets without requiring further assembly or the specialized targeting of the 16S rRNA marker gene. In addition, MEGAN can do statistical analysis and provide graphical output to compare various datasets [103]. Another noteworthy tool is MetaPhlAn, which is an analytic tool specifically designed for Metagenomic Phylogenetic analytics (Met-aPhlAn). This tool is specially engineered to expedite taxonomic classifications by eliminating duplicate sequences. This programme presents a distinct method for taxonomic assignments, which may differ from the methods provided by RAMMCAP and MEGAN [107, 109].

Short readings, referred to as DNA sequence fragments and must be assembled into larger sequences known as contigs. The length of these contigs is comparable to that of a gene, enabling the possibility of annotating their function. To assemble these contigs, one can use specialised software tools like MetaVelvet and SOAPdenovo2 [104, 110]. These techniques facilitate the merging of small reads into larger contigs, hence enabling the investigation of their function. Aside from assembly, certain software tools such as MOCAT can concurrently assemble the short reads into contigs and annotate the genes included within them. This procedure involves doing quality assurance assessments and predicting the genes from the contigs [105]. To explore the functional characteristics of the metagenomic reads, one can employ either the predicted genes derived from the assembled contigs or the raw sequence reads with a greater read length. Subsequently, these anticipated genes or unprocessed sequence reads can undergo functional annotation [111]. The Kyoto Encyclopaedia of Genes and Genomes (KEGG) offers a comprehensive database that categorises genes into KEGG enzymes, pathways, and orthologs, serving this specific objective. This organisation facilitates a more profound comprehension of the metabolic capacity of the microbial population being investigated [111].

Several methods have been created to simplify and automate the assembly and annotation of metagenomic data. An example of such a pipeline is SmashCommunity, which provides a user-friendly graphical interface for the examination of metagenomic data. Another instance is the Microbiome Project Unified Metabolic Study Network (HUMANN), which offers a comparable graphical user interface (GUI) technique for study [112]. Furthermore, the FANTOM pipeline is also accessible for automating the processes of metagenome assembly and annotation. These

pipelines are useful tools for researchers since they streamline and accelerate the processing of metagenomic data [113].

4.4 Metatranscriptomics

Metatranscriptomics can provide valuable insights into the functional activity of the gut microbiota in patients with T2D. By analyzing the RNA transcripts of the microbial community, researchers can understand which genes are actively expressed, providing a more dynamic view of the microbiome's functional potential [114]. In T2D, the gut microbiota composition and function are often altered, leading to dysbiosis. Metatranscriptomic analysis can help identify specific microbial genes and pathways that are differentially expressed in diabetic patients compared to healthy individuals. This information can shed light on the metabolic processes, such as carbohydrate metabolism or inflammation, that may be influenced by the microbiota in diabetes [115]. Next-generation sequencing has revolutionised the field of Metatranscriptomics, enabling researchers to analyse microbial communities directly and investigate gene expression to determine the function of organisms [116]. The application of meta-transcriptomics in studying diabetic disease revealed that the genes involved in carbohydrate metabolism in Muribaculaceae were upregulated, whereas the genes responsible for vitamin and cofactor metabolism, as well as amino acid metabolism, were upregulated in Deferribacteraceae [117]. The RNA extracted from faeces represents the processes of gastrointestinal activity, given there are fast fluctuations in the mRNA transcript pool. The cause of this phenomenon may not be attributed to challenging sampling circumstances, but rather to the fast fluctuations in the mRNA transcript pool [118].

4.5 Metagenomic assembly and gene prediction

To begin the process of functionally characterising metagenomic whole genome sequencing (mWGS) samples, the initial stage often involves performing de novo metagenome assembly, followed by gene annotation. Metagenomic assemblies often yield several genomic fragments, referred to as contigs, rather than whole genomes. By consolidating short fragments into contiguous sequences, predictions for both entire and partial gene sequences may be computed, extracted, and contrasted with established gene databases [119].

Estimating genes from metagenomic assemblies has many extra challenges compared to predicting genes in individual genomes. An extremely fragmented assembly is more prone to have incomplete annotations as a result of its fragmented nature. Furthermore, metagenomes may lack the shared traits that bring genes together within a single genome and aid in their distinction. Several pre-existing technologies are available specifically designed to assist in the annotation of meta-genomes. While certain tools like MetaGeneMark and Prodigal are capable of handling this task, their accuracy in predicting new genes inside the metagenome's dark matter may be limited

[120, 121]. With the significant growth of prokaryote genome annotations in recent years, there is a need for new techniques to enhance the accuracy of gene annotations derived from metagenome assemblies [122].

4.6 Functional genome annotation

Functional reference databases carefully select groups of homologous genes. A KEGG ORTHOLOGY (KO) database or EggNOG orthologous group can demonstrate the shared function of these groupings. Functional annotation can be accomplished by aligning query sequences with specific reference sequences or categorizing them accordingly. HMM profiles demonstrate significant efficacy in the realm of sequence characteristics [123]. These profiles, which exhibit substantial similarities to group features and very modest similarities to their constituents, have the potential to enhance metagenomic functional annotation [124]. The functional annotation tools and workflow are described in Figure 4.

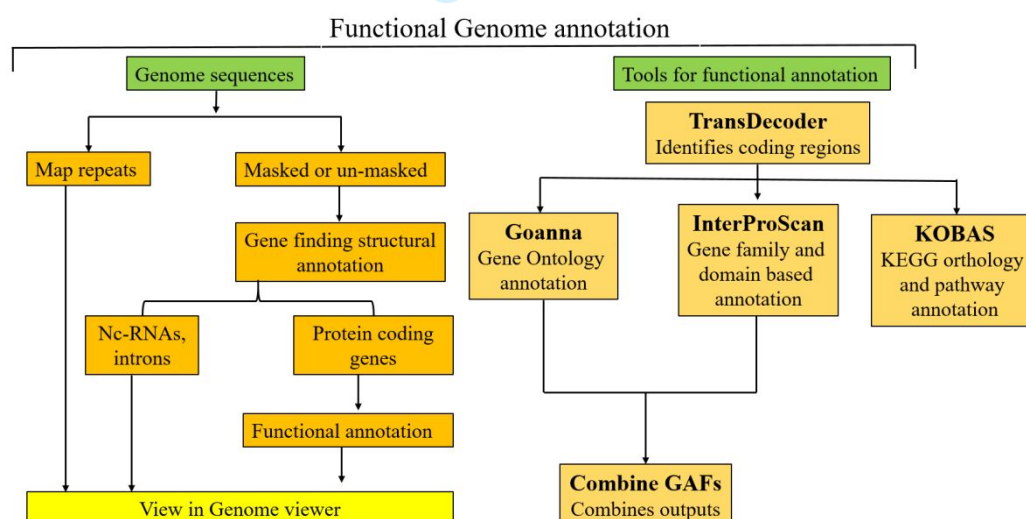


Figure 4. Schematic representation of functional genome annotation

KofamKOALA, a specialised server supplied by the Kyoto Encyclopaedia of Genes and Genomes (KEGG), utilises an approach based on hidden Markov models (HMM). This server can produce KO classifications for translated metagenome annotations [125]. In addition, Kanehisa et al. [126] suggest aligning query sequences with KEGG GENES and subsequently assigning them to KO groups based on the obtained findings. The eggNOG-mapper is a software tool created by the curators of the evolutionary genealogy of genes: Non-supervised Orthologous Groups (eggNOG) to match query sequences with eggNOG orthologous groups. Classification queries with very big datasets may have performance issues. The DIAMOND aligner may also use this approach to compare queries with seed sequences for each set of eggNOG orthologs [117].

The above-mentioned, annotation systems are generally meant to identify gene sequences either in their entirety or in part. Furthermore, the HUMAnN technique can directly measure functions, in addition to unprocessed data. HUMAnN3 is an enhanced iteration of the database that combines

UniProtKB sequences and functional annotations with GenBank genome sequences and taxonomies. A hierarchical search involves the use of nucleotide and translated alignments to increase the probability of accurately classifying reads. HUMAnN enables quantitative examination of taxonomic contributions in databases like MetaCyc and KEGG by preserving information regarding the source genomes [128].

4.6.1 Exploring the relationship between genetic information and biological mechanisms

Insight into a metagenome may be obtained by examining the quantity of orthologous groups that have functional annotations. Discovering gene clusters or pathways that accurately represent the desired activities of the microbiome is more probable than identifying clusters of genes or pathways that primarily indicate broader biological processes. All of these genes participate in the production of secondary bile acids. Quantifying metagenomic pathways is a significant challenge. Various taxonomic groups may possess genes associated with distinct pathways, and many genes have a role in numerous pathways. Due to this constraint, there is a reduced availability of techniques for determining the abundance of pathways in metagenomic data. HUMAnN3 is used to accurately link gene annotations to MetaCyc reactions and higher-order pathways [129].

There are two approaches to ascertain the quantity of particular routes. An approach involves analysing the relative abundance of all genes or orthologous groups that contribute to the pathway as a whole. This is very advantageous for measuring KEGG Orthologs. Another approach is gene set analysis (GSA), which provides a single quantitative estimate of abundance. The GSA methodologies employed in metagenomics are analogous to those utilised in other genomics fields. To identify an overrepresentation of a certain set among all genes that exhibit substantial differential abundance between distinct situations, search. On the other hand, search for the enhancement of a group inside a pre-established categorization of all genes, depending on the extent and orientation of their relative abundance across different situations. It is important to highlight that Huang et al. [130] offer a further understanding on this subject.

The inclusion of GSA and route abundance metrics is undeniably essential for the analysis of major biological processes using metagenome-wide association studies (mWGS) data. However, effectively representing the diverse range of microbial genes, which vary in both function and taxonomy, as separate pathways may be a difficult task. Therefore, it is imperative for researchers to thoroughly analyse the particular genes that contribute to the observed patterns at elevated functional levels. Hence, it is imperative for the scientific community to methodically extract meaningful insights from these intricate databases [128, 130].

5.0 Some advanced methods for gut microbial analysis

5.1 Spatially resolved microbiomes

To gain a comprehensive understanding of metagenomic samples of the microbiome, it is crucial to consider the environmental context. This is especially significant in the gut, where even minor microbiome structure fluctuations can be detected between the gastrointestinal lumen and the intestinal mucosal barrier [131]. A new technique known as High Phylogenetic Resolution Fluorescent in situ Hybridization enables the identification and location of various microorganisms [132]. The method involves using a probe that targets the 16S rRNA molecule and a fluorescently labelled second probe. A spectral barcode that is specific to a certain type of microbe can be generated by employing multiple probe markers. This system can produce more than 1000 distinct fluorophore pairs, making it possible to target different microbe types in a single assay. Using single-cell resolution imaging and a machine-learning-based classifier, the spectra emitted by each cell can be decoded, thereby allowing the identification and location of tagged cells in a multi-taxonomic bacterial community [133]. Researchers can use quantitative spatial mapping to investigate the connections between microorganisms and their surroundings. In this research, the authors meticulously mapped the microbiota in the colon tissue of mice utilizing HiPR-FISH. They scrutinized the spatial connections among various bacterial taxa. However, the authors discovered that the utilization of antibiotics caused a disturbance in the association between bacterial partners, such as *Oscillibacter* and *Veillonella*. Ultimately, this novel approach enables the examination of intricate microbial communities at the individual cell level while also considering their spatial orientation [133, 134].

5.2 Single-cell metagenomics

The study of host genomics has been dramatically enhanced by single-cell sequencing, which enables researchers to analyze specific cell populations that would have been overlooked with conventional transcriptomic techniques. In microbiome research, similar methods necessitate the physical separation of microorganisms into groups of single cells or small clusters of cells known as mini-metagenomes [135]. Flow-sorting cells, gel microdroplet cultivation, single droplet multiple displacement, and microfluidic hydrogels have all been used to achieve this [136, 137]. Chijiwa et al. [137] used gel microfluidic hydrogels to isolate individual cells, enzymatically lysed their DNA, and amplified their genomes within each hydrogel, creating Single-Amplified Genomes (SAGs). Despite the limited completeness of SAG genomes, the authors generated numerous high-quality draft genomes that could be used to construct complete or composite genomes of SAG strains [123]. This methodology offers a significant benefit in enabling the creation of genomic sequences for uncommon taxa without requiring a substantial boost in sequencing depth. Single amplified genomes (SAGs) necessitate only tens of thousands of reads,

making it feasible to sequence numerous genomes simultaneously and employ mini-metagenomic sequencing on less costly platforms, such as the Illumina MiSeq [137].

6.0 Conclusion and future prospects

Metagenomics methods are considered for the identification and characterization of gut microbiota in the human gut. Target metagenome sequencing and whole genome sequencing provide structural and functional understanding of the gut microbial community. This undertaking involves the use of whole genome sequencing (WGS) and potentially the cultivation of individual organisms. Moreover, there are many undiscovered ecosystems present in the gastrointestinal tract, specifically regarding the variety of genetic material from microorganisms. In the future, there will be a growing number of research focused on understanding the functions of the microbiome, and RNA sequencing (RNA-seq) may have a significant impact in this regard. Yet, the difficulty resides in the process of producing metatranscriptomes by preparing high-quality sample RNAs for sequencing.

The relationship between gut microbiota and human illnesses has successfully obscured the differentiation between infectious and non-infectious diseases. While the alteration of the microbiota has significant therapeutic possibilities, there is a requirement for the advancement of techniques that facilitate the manipulation of specific bacteria within a community and enable precise therapy, such as designer probiotics. Additionally, there is a pressing need for inventive methods to create extensive networks of associations inside the gut ecosystem. These networks would aid in the creation of global models that accurately represent the dynamics of the gut ecosystem. These models would therefore allow for the anticipation of disturbances in the gut and eventually assist in therapeutic actions.

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